Mosquitoes spatial model

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1 Model Equations

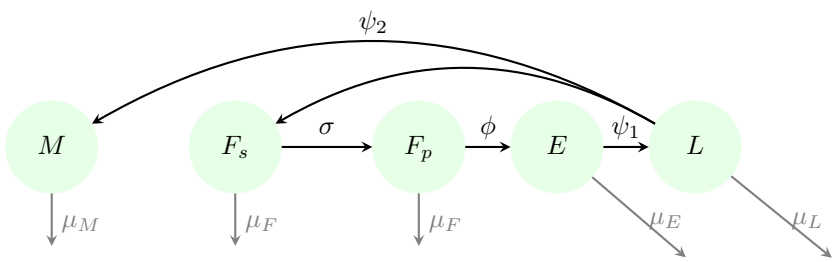

Figure 1: General model: Single Species

Table 1: Variables

Symbol	Description
E	Egg stage
L	Larvae/pupae stage
M	Male mosquitoes
F_s	Unmated female mosquitoes
F_p	Pregnant female mosquitoes

Table 2: Parameters

Symbol	Description	Units
Parameters for 5-ODE system		
b_F	Female birth probability	
b_M	Male birth probability	
σ	Per capita mating rate	
ϕ	Per capita egg laying rate F_p	
ψ_1	Per capita development rate from egg into larvae/pupae stage	
ψ_2	Per capita development rate from larvae/pupae to adult stage	
μ_E	Death rate for E	
μ_L	Death rate for L	
μ_F	Death rate for F_s or F_p	
μ_M	Death rate for M	
K	Carrying capacity for L_i, $i = a, b$	
Parameters for 4-ODE system		
$\phi' = \dfrac{\phi\sigma}{\mu_F + \sigma}$	Per capita egg laying rate F	

1.1 5-ODE model

1.1.1 General Model: Single Species

$$
\begin{aligned}
\frac{dE}{dt} &= \phi F_p - (\mu_E + \psi_1)E \\
\frac{dL}{dt} &= \psi_1 E\left(1 - \frac{L}{K}\right) - (\mu_L + \psi_2)\,L \\
\frac{dM}{dt} &= \psi_2 b_M L - \mu_M M \\
\frac{dF_s}{dt} &= \psi_2 b_F L - (\sigma + \mu_F)F_s \\
\frac{dF_p}{dt} &= \sigma F_s - \mu_F F_p
\end{aligned}
$$

Equilibrium states of the 5-ODE General Model: Disease-free equilibrium (DFE),

$EE^0 = (E^0, L^0, M^0, F_s^0, F_p^0)$

$$
\begin{aligned}
E^0 &= b_F \frac{\phi}{\mu_F} \frac{\sigma}{\sigma + \mu_F} \frac{\psi_2}{\psi_1 + \mu_E} L^0 \\
L^0 &= K \left(1 - \frac{1}{\mathbb{G}_0}\right) \\
M^0 &= b_M \frac{\psi_2}{\mu_M} L^0 \\
F_s^0 &= b_F \frac{\psi_2}{\sigma + \mu_F} L^0 \\
F_p^0 &= b_F \frac{\sigma}{\mu_F} \frac{\psi_2}{\sigma + \mu_F} L^0
\end{aligned}
$$

Next generation number:

$$
\mathbb{G}_0^{(5)} = b_F \frac{\psi_1}{\mu_E + \psi_1} \frac{\psi_2}{\mu_L + \psi_2} \frac{\sigma}{\sigma + \mu_F} \frac{\phi}{\mu_F}
$$

4-ODE Model: Combining F_s and F_p (Non pregnant and pregnant females)

$F = F_s + F_p$. The new egg-laying rate is defined by matching the basic next generation numbers of the 5-ODE and the 4-ODE model,

$$
\mathbb{G}_0^{(4)} = b_F \frac{\psi_1}{\mu_E + \psi_1} \frac{\psi_2}{\mu_L + \psi_2} \frac{\phi'}{\mu_F}
$$

where

$$
\phi' = \frac{\phi\sigma}{\mu_F + \sigma}.
$$

The reduced 4-ODE model is

$$
\begin{aligned}
\frac{dE}{dt} &= \phi' F - (\mu_E + \psi_1) E \\
\frac{dL}{dt} &= \psi_1 E \left(1 - \frac{L}{K}\right) - (\mu_L + \psi_2) L \\
\frac{dM}{dt} &= \psi_2 b_M L - \mu_M M \\
\frac{dF}{dt} &= \psi_2 b_F L - \mu_F F
\end{aligned}
$$

3-ODE Model: Removing the males

The reduced 3-ODE model is

$$\begin{aligned}
\frac{dE}{dt} &= \phi' F - (\mu_E + \psi_1)E \\
\frac{dL}{dt} &= \psi_1 E \left(1 - \frac{L}{K}\right) - (\mu_L + \psi_2)\, L \\
\frac{dF}{dt} &= \psi_2 b_F L - \mu_F F
\end{aligned}$$

$$\mathbb{G}_0^{(3)} = b_F \frac{\psi_1}{\mu_E + \psi_1} \frac{\psi_2}{\mu_L + \psi_2} \frac{\phi'}{\mu_F}$$

1.2 Two species model:

Aedes Aegypti:

$$\frac{dE_a}{dt} = \phi'_a F_a - (\mu_{E_a} + \psi_{1_a})E_a \tag{1}$$

$$\frac{dL_a}{dt} = \psi_{1_a} E_a \left(1 - \frac{L_a + L_b}{K}\right) - (\mu_{L_a} + \psi_{2_a})\, L_a \tag{2}$$

$$\frac{dF_a}{dt} = \psi_{2_a} b_{F_a} L_a - \mu_{F_a} F_a \tag{3}$$

$$\tag{4}$$

Aedes Albopictus:

$$\frac{dE_b}{dt} = \phi'_b F_b - (\mu_{E_b} + \psi_{1_b})E_b \tag{5}$$

$$\frac{dL_b}{dt} = \psi_{1_b} E_b \left(1 - \frac{L_a + L_b}{K}\right) - (\mu_{L_b} + \psi_{2_b})\, L_b \tag{6}$$

$$\frac{dF_b}{dt} = \psi_{2_b} b_{F_b} L_b - \mu_{F_b} F_b \tag{7}$$

$$\tag{8}$$

$$\frac{dE_b}{dt} = \phi'_b(T) F_b - (\mu_{E_b}(T) + \psi_{1_b}(T))E_b \tag{9}$$

$$\frac{dL_b}{dt} = \psi_{1_b}(T) E_b \left(1 - \frac{L_a + L_b}{K(R)}\right) - (\mu_{L_b}(T) + \psi_{2_b}(T) + c_1(R)L_b + c_2(R)L_a)\, L_b \tag{10}$$

$$\frac{dF_b}{dt} = \psi_{2_b}(T) b_{F_b}(T) L_b - \mu_{F_b}(T) F_b \tag{11}$$

$$\tag{12}$$

We are considering two species of mosquitoes in the model; species A (Aedes Aegypti), which has been present in the area for years (*), and species B (Aedes Albopictus), which was first captured in the area in 2018 (*). Therefore, there are two types of equilibrium points corresponding to: (DFE_A) Only species A is present, (EE) species A and B are present. The DFE_A (**Single species equilibrium**) is found by setting $E_b = L_b = F_b = 0$, then $DFE_A = \left(E_a^0, L_a^0, F_a^0, 0, 0, 0\right)$:

$$
\begin{aligned}
E_a^0 &= b_{F_a} \frac{\phi_a'}{\mu_{E_a} + \psi_{1_a}} \frac{\psi_{2_a}}{\mu_{F_a}} L_a^0 \\
L_a^0 &= K \left(1 - \frac{1}{\mathbb{G}_{0_a}^{(3)}}\right) \\
F_a^0 &= b_{F_a} \frac{\psi_{2_a}}{\mu_{F_a}} L_a^0
\end{aligned}
$$

The next generation number for species A,

$$
\mathbb{G}_{0_a}^{(3)} = b_{F_a} \frac{\psi_{1_a}}{\mu_{E_a} + \psi_{1_a}} \frac{\psi_{2_a}}{\mu_{L_a} + \psi_{2_a}} \frac{\phi_a'}{\mu_{F_a}}
$$

represents the number of eggs that one egg of species A can generate within one life cycle of a mosquito. In a wild mosquito population, $\mathbb{G}_{0_a}^{(3)} > 1$ is the essential condition that guarantees the persistence of the natural population and, therefore, we assume that $\mathbb{G}_{0_a}^{(3)} > 1$.

1.2.1 Stability of the DFE: The basic reproductive number $\mathbb{R}_0$

The basic reproductive number serves as an invasion threshold condition and determines the invasion of species B in a location where species A is endemic. Without loss of generality, we define the $DFE = DFE_A = \left(E_a^0, L_a^0, F_a^0, 0, 0, 0\right)$ where only species A is present. Thus, the threshold for species B invasion is given by the basic reproduction number of the two-species system. We derive this number using the next-generation method. The "infected" compartments of the system correspond to the invading species b: $\vec{X} = (0, 0, 0, E_b, L_b, F_b)$. $\mathcal{F}$ is the rate of new infections and $\mathcal{V}$, the rates of transitions.

$$
\frac{d\vec{X}}{dt} = \frac{d}{dt}\begin{pmatrix} E_b \\ L_b \\ F_b \end{pmatrix} = \begin{pmatrix} \phi_b' F_b \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} (\mu_{E_b} + \psi_{1_b}) E_b \\ \psi_{1_b} E_b \left(1 - \dfrac{L_a + L_b}{K}\right) + (\mu_{L_b} + \psi_{2_b}) L_b \\ -\psi_{2_b} b_{F_b} L_b + \mu_{F_b} F_b \end{pmatrix} =: \mathcal{F} - \mathcal{V}
$$

Table 3: Parameters: Aedes Aegypti

Symbol	Value	Description
Parameters for 5-ODE system		
b_{F_a}	0.5	Female birth probability
b_{M_a}	0.5	Male birth probability
σ		Per capita mating rate
ϕ_a		Per capita egg laying rate F_p
ψ_1	1/16	Per capita development rate from egg into larvae/pupae stage (Lita data)
ψ_2	1/2	Per capita development rate from larvae/pupae to adult stage (Lita data)
μ_E	0.0156	Death rate for E (calculated)
μ_L	0.0682	Death rate for L (calculated)
μ_F	1/17.5	Death rate for F_s or F_p (ZQ)
μ_M	1/10.5	Death rate for M (ZQ)
K		Carrying capacity for L_i, $i = a, b$
Parameters for 4-ODE system		
$\phi'_a = \dfrac{\phi\sigma}{\mu_F + \sigma}$	3.54	Per capita egg laying rate F (Lita data)

Table 4: Parameters: Aedes Aegypti

Symbol	Value	Sources
Parameters for 5-ODE system		
b_{F_a}	0.5	
b_{M_a}	0.5	
σ	1	
ϕ_a		Mordecai $(5.7852, 6.1156)$ and Yang 6.3 at 25 C
ψ_1		$(1/8.7489, 1/9.1575)$
ψ_2		Mordecai and Yang? (confirm)
μ_E		Mordecai $(8.2531e - 04, 0.0017)$
μ_L		Yang (0.057)
μ_F		Mordecai $(1/29.1545, 1/29.3255)$ and Yang(1/30)
K		Carrying capacity for L_i, $i = a, b$
Parameters for 4-ODE system		

The Jacobian matrices of $\mathcal{F}$ and $\mathcal{V}$ at the DFE:

$$J_{\mathcal{F}} = \frac{d\mathcal{F}}{dt} = \begin{pmatrix} 0 & 0 & \phi_b \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad J_{\mathcal{V}} = \frac{d\mathcal{V}}{dt} = \begin{pmatrix} \mu_{E_b} + \psi_{1_b} & 0 & 0 \\ -\left(1 - \frac{L_{a_0}}{K}\right)\psi_{1_b} & \mu_{L_b} + \psi_{2_b} & 0 \\ 0 & b_{F_b}\psi_{2_b} & \mu_{F_b} \end{pmatrix}$$

The basic reproductive number is calculated as the spectral radius of the next generation matrix $J_{\mathcal{F}} J_{\mathcal{V}}^{-1}$,

$$\mathbb{R}_0 = \frac{\mathbb{G}_{0_b}}{\mathbb{G}_{0_a}}$$

2 Parameters

2.1 Aedes aegypti

Parameters from Yang 2011 [2]:

1. ϕ : per-capita oviposition rate.
2. $\mu_f = \mu_F$: death rate of adult female mosquito.
3. $\mu_q = \mu_L$: death rate of the aquatic stage of the mosquito.
4. $\pi_q = \psi_{2_a}$: emergence rate of pupae into adults.

Parameters in our paper we obtain from Yang and Farnessi:

- **Per capita reproduction rate (ϕ'):** We use ϕ, μ_F from Table 1 (Yang) and $\sigma = 1$ (Per capita mating rate) to compute $\phi' = \dfrac{\phi\sigma}{\mu_F + \sigma}$.
- **Embryonic Stage Death Rate (μ_{E_a}):** We determine the death rate of the embryonic stage (μ_{E_a}) based on Farnessi's data (Table on page 125) [1]. We fit column 1 of the table to derive the polynomial representing the temperature-dependent completion of embryogenesis ($CE = (1/\mu_E) * 24\text{hours}$) from egg to the first larval stage (L_1). The value of μ_{E_a} is then obtained from this formula.
- **Per Capita Development Rate from Egg to Larvae/Pupae Stage:** Utilizing data from Farnessi's Table on page 125 [1], we estimate ψ_1. By fitting the Eclosion % (EP) column of the table, we derive ψ_{1_a} using the formula
$EP = \dfrac{\psi_{1_a}}{\psi_{1_a} + \mu_{E_a}} \rightarrow \psi_{1_a} = \dfrac{\mu_{E_a} EP}{1 - EP}$.
- The following parameters are used directly from Yang's paper:
$\mu_f = \mu_F$, $\mu_q = \mu_L$ and ψ_{2_a}.

2.2 Aedes Albopictus

Parameters are from Mordecai [3] appendix S1 Text, Fig. A.

- μ_{F_b}: mosquito lifespan (Adult Lifespan Fig. 1)
- ψ_{2_b} : We obtain ψ_2 from the development rate curve from [3]: $MDR = \dfrac{1}{\psi_{2_b}}$
- ϕ_b: Per capita egg laying rate F_p (Fecundity Fig. A, Supplementary material). The units of the data on the Supplementary material are in [Eggs per female per cycle], so to transform it to [Eggs per female], we divide it by the duration of the gonotrophic cycle. To obtain the length of the gonotrophic cycle, we use the biting rate plot in Fig. A, Supplementary material, because *"gonotrophic cycle duration (GCD, which we assumed was equal to the inverse of the biting rate)"*.
- μ_{L_b} : We obtain μ_{L_b} from the plot of the Egg-to-Adult survival $= \dfrac{\psi_{2_b}}{\psi_{2_b} + \mu_{L_b}}$ in Fig. A, Supplementary material.

- $1/\psi_{1_b}$: We use Table 2, column ST since the area is subtropical [4] and fit a polynomial to obtain the temperature-dependent curve.
- μ_{E_b} : Use ST in column Egg-L1 from Table 1 in [4] to obtain $\psi_{1_b}/(\psi_{1_b}+\mu_{E_b})$ =survival rate from Egg to larval stage L1.

For temperature function, check nonlinear regression with cosine functions in White Paper 2015.

2.2.1 Temperature

2018 - 2019 Temperature data:

```
Location: /Dropbox/Spatial Mosquito project/Data/temperatura Lita/temp data
final/SIT_Tpr_HOBO_Ovitramp_20180108-20191219_LITA-
Cachaco-Guadual_Consolid_KL_MB copy.xlsx
```

Water temperature of the ovitraps obtained from the dataloggers (brand: HOBO) placed inside the ovitraps between January 8, 2018, and December 19, 2019.
2021 - 2023 Temperature data:

```
Location: /Users/joanponce/Library/CloudStorage/Dropbox/Spatial Mosquito
project/Data/temperatura Lita/temp data final/SIT_Tpr-HR_ambt_20210722-
20230613_DATALOGGERS_LITA-CACHACO_XAG copy.xlsx
```

Environmental temperature was obtained from the dataloggers (brand: HOBO) placed inside houses between 22 July 2021 and 13 June 2023.

References

[1] Luana Cristina Farnesi et al. "Embryonic development of Aedes aegypti (Diptera: Culicidae): influence of different constant temperatures". In: Memórias do Instituto Oswaldo Cruz 104 (2009), pp. 124–126.

[2] Hyun Mo Yang et al. "Follow up estimation of Aedes aegypti entomological parameters and mathematical modellings". In: Biosystems 103.3 (2011), pp. 360–371.

[3] Erin A Mordecai et al. "Detecting the impact of temperature on transmission of Zika, dengue, and chikungunya using mechanistic models". In: PLoS neglected tropical diseases 11.4 (2017), e0005568.

[4] Giovanni Marini et al. "Influence of temperature on the life-cycle dynamics of Aedes albopictus population established at temperate latitudes: A laboratory experiment". In: Insects 11.11 (2020), p. 808.